Land Use and Land Cover Area Estimates From Class Membership Probability of a Random Forest Classification

Marcio H. R. Sales, Sytze de Bruin, Carlos Souza, Jr., and Martin Herold

***Abstract*— Estimates of the area of land cover classes or land change are frequently calculated from land cover classification maps by counting the pixels labeled as each class in the map. This procedure is known to produce biased estimates of area for many widely used classification algorithms, including random forests. Poststratification estimation using the mapped classes as strata has been proposed to obtain unbiased estimates of the class areas. Still, the method requires additional sampling units, which may not be available or be the most efficient method depending on the application. Alternatively, consistent estimates of class areas can be obtained using class membership probabilities estimates from a random forest classification. This article demonstrates that, for a large sample and proper set of explanatory variables, the error of the predicted class membership probabilities obtained from a random forest classification converges to zero. Therefore, the expected class areas calculated from these probabilities converge to the population class areas. On average, the relative error of the expected class proportions computed by class membership probabilities from a random forests model was 40% points lower than the proportions estimated by pixel counting. Our proposed approach is also comparable to the area-adjusted method, which is currently considered the best practice by the remote sensing community. We recommend that class probability estimates area always retained and used for calculating expected class areas or area proportions based on our results. Our method reduces bias compared to statistics calculated by pixel counting and circumvents the need for poststratification area estimates under certain conditions.**

***Index Terms*— Area estimation bias, computational infrastructure, optical data, random forest and geographic information system (GIS), vegetation and land surface.**

I. Introduction

CHANGES in land use and land cover (LCLU) are among the main drivers of current changes in Earth's ecosystems and the global climate [1]. Because of the harms caused by these changes and the benefits of resources and services that land use provides to humanity contrast [2], mapping and monitoring LCLU are crucial for understanding the evolution of these changes on Earth's surface and to sustainable policy-making [3]–[5]. LCLU mapping applications range from land management [6] to ecology [7] and from ecosystem sustainability to public health surveillance [8]. Given the detailed insight that LULC mapping products provide about the landscape, it used to be widely accepted that such products yield accurate estimates of LULC class areas directly [9]. Unfortunately, land cover class areas derived from land cover maps' pixel counts are almost always biased. This is the case for many commonly used classification algorithms [10]. Methods based on reflected radiation's physical properties, such as spectral unmixing (SU) [11], can be directly used to estimate areas if well-calibrated. However, random forests [11] are becoming increasingly popular for their ease of use and superior spatial accuracy [12]–[14]. Random forest models producing land cover maps use satellite information to predict each pixel's land cover class in a map. This happens because the outcome of random forest and many other classification algorithms when evaluated on spectral measurements in a pixel is not a class label but a vector of class membership probabilities. These class membership probabilities reflect the modeled "chance" that the pixel being evaluated pertains to each class, which are afterward collapsed into a single label, which is necessary for displaying results in a map. However, the area of a land cover class over a region is commonly estimated by summing the areas of all pixels labeled as that class. This approach is referred to as "pixel-counting" area estimates [9], [15], [16]. Despite the spatial resolution of the land cover map, if some classes are much more abundant than others, it is common that area estimates calculated based on the counting pixels overestimate abundant classes and underestimate scarce classes [17], [18]. This happens due to the data loss incurred by collapsing the probabilities, which is done by changing the membership probabilities to one for the highest probability class and zero for all others. Pixel-counting area estimates can be corrected by poststratification using extra reference data [19], [20]. However, by statistical sampling theory [21], achieving adequate precision for area estimates for small classes may require a large investment in reference data collection, even if poststratification is implied.

Alternatively, instead of LCLU class label assignments, the class membership probabilities can be used to calculate class areas' estimates. It has been proved that random forest class probabilities are consistent estimates of the true probabilities [10]. As a result, providing that a sufficiently large random sample is used for calibration of the random

Manuscript received October 15, 2020; revised February 8, 2021; accepted March 28, 2021. Date of publication June 9, 2021; date of current version January 12, 2022. This work was supported by the Porticus Latinamerica. *(Corresponding author: Marcio H. R. Sales.)*

Marcio H. R. Sales, Sytze de Bruin, and Martin Herold are with the Laboratory of Geo-Information Science and Remote Sensing, Wageningen University & Research, 6708 PB Wageningen, The Netherlands (e-mail: marcio.ribeirosales@wur.nl).

Carlos Souza, Jr. is with the Instituto do Homem e Meio Ambiente da Amazônia (Imazon), Belém 66055-200, Brazil.

Digital Object Identifier 10.1109/TGRS.2021.3080083

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.

forest model and assuming that the used covariates have good predictive power, the total area of each land cover class can be estimated by summing of the pixels' corresponding class membership probabilities multiplied by the pixel area. This seems to be commonly overlooked in remote sensing applications because simple pixel counting is still used to report land cover class areas [22]–[26].

This study demonstrates that class membership probabilities from random forest models provide consistent and more cost-efficient estimates of LULC class areas over long periods. This is accomplished by comparing area estimates obtained by pixel counting of land cover classifications with the same estimates obtained using the class membership probability estimates from a trained random forest algorithm. The work is motivated by the recent availability of a large reference data set collected in the Brazilian Amazon for a time series from 1990 to 2018 for various land cover classes.

We start with a brief review of the random forest algorithm and explain the rationale of using the membership probabilities to estimate class areas. We then move on to describe the design of our analysis. We show the formulas for pixel-counting, poststratification, and probability-based LCLU class area estimates and visually and quantitatively compare their results. We close the article discussing the implications and providing recommendations on when the probabilities and poststratification estimates are best used.

II. Materials and Methods

A. Random Forest Classification and Class Area Estimates

Random forest is an ensemble learning algorithm [12], [13], [27], [28] based on classification or regression trees. Each tree of the ensemble is grown with a different bootstrap sample (or subsample) of the original data set and a random selection of a fixed number of variables extracted from the entire feature space to define each tree's optimal node splits [12]. For prediction, features (or variables) of each pixel are evaluated by each tree and assigned a class label in the ensemble. The predicted pixel class is decided by the majority voting over the trees in the ensemble, and a map of land cover classes is produced after predicting at all pixels. Alternatively, instead of voting, the proportion of trees voting for each class can be retained. These can be considered as an estimate of the conditional probability that the pixel belongs to that class, given its spectral signature [10].

When using the class membership probabilities, the expected value of the area of class k is be calculated by

$$E(S_k) = \sum_{i=1}^{N} p_i(\text{class} = k) \times s_i \tag{1}$$

and its variance is given by

$$\text{Var}(S_k) = s^2 \times \left(\sum_{i}^{N} \text{var}(p_i(\text{class} = k)) + 2 \sum_{i,j,i \neq j} \text{Cov}\big(p_i(\text{class}{=}k),\, p_j(\text{class}{=}k)\big) \right) \tag{2}$$

where k is the class index, $p_i(\text{class} = k)$ is the class membership probability of class k for pixel i, s_i is the area of pixel i, and $s = \sum_N s_i$. Therefore, the variance would score nearly 0 for classes with probabilities mostly close to 0 or 1 and nearly $0.25 \times (\sum s_i)^2$ for classes with most probabilities close to 0.5 or if pixels are fully correlated.

Baeza and Paruelo [26] show that random forests are a form of weighted layered nearest neighbors' model and generate consistent estimates. This implies that predicted probabilities converge to the class membership probabilities when the sample size fits the model increases to infinity. Therefore, the expected class areas of (1) converge to the true class areas with increasing sample size. In contrast, estimates of the total area using discrete label predictions are biased because sparser classes are unlikely to attain the highest class probability used for a discrete class assignment. To show this empirically, we compare two estimates using the reference data set. Estimates are calculated using the random forest algorithm model on the Landsats 5, 7, and 8 data available in Google Earth Engine [30]. For this work, we have opted to estimate membership probabilities by the proportion of the number of tree votes for each class, i.e., classification through random forests. However, consistency is also granted and could be even faster if regression forest was used [10].

B. Study Area and Reference Data

The reference data used in this study come from the Mapbiomas project, "a collaborative network of Brazilian biomes, land use, remote sensing, geographic information system (GIS), and computer science experts. The MapBiomas project relies on the Google Earth Engine platform and its cloud processing and automated classifiers capabilities to generate Brazil's annual LCLU time series" [31]. The Mapbiomas' reference data set comprises tens of thousands of sampling pixels. The project collected colocated classifications of land cover by careful visual interpretation of Landsat data for entire Brazil from 1985 to 2018. Our data set used the subset of all *25* 254 pixels located within the Amazon Biome boundaries (see Fig. 1). The data set included, by 2018, 28 LULC classes in three taxonomic levels of detail (see Table I), 13 of which were present in the reference data set over the Amazon. More information about this data set is found in the Mapbiomas Algorithm Theoretical Base Document (ATBD) [32]. Mapbiomas reports that the accuracy level of the reference visual classifications is above 90%. We used the second classification level (level 2 labels) of the legend to ensure that enough data were available to train the classifier and make predictions while keeping a relatively large number of classes, including both abundant and sparse ones.

From the MapBiomas' ATBD, the actual sampling design was stratified by classes of slope ranges. However, low slope variability in the region resulted in similar sampling weights, and thus, the analysis was conducted as if the pixels' locations have been selected by simple random sampling.

Most of the pixels (~79% on average over the years) in our data set were in forests (see Table II). For some of the classes,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.

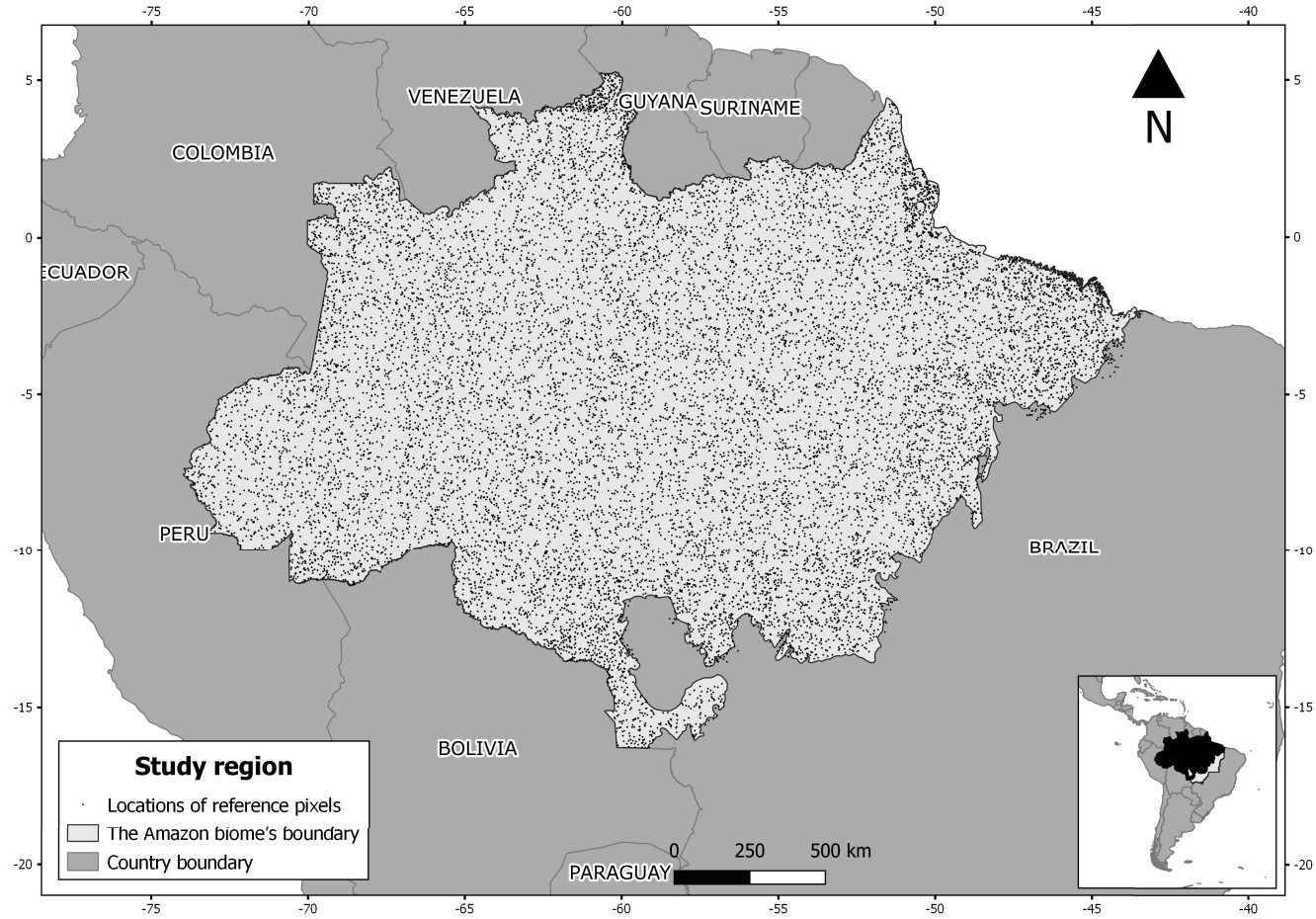


Fig. 1. Study region. Dots show the location of the subset of Mapbiomas' reference data set covering the Brazilian Amazon.

the average annual number of pixels is below 20. Due to small sample sizes, predictions for "other nonvegetated areas" and other small classes have high variance. Still, we decided to assess them separately to compare results with the estimates obtained for the abundant classes.

C. Remote Sensing Data

We used Google Earth Engine to extract Landsats 5, 7, and 8 reflectance data for all pixels in our reference data set over the study region from 1990 to 2018. The "pixel quality" band of each product was used to mask cloud and other types of bad pixels for each sensor. Because we only knew the year of the visual interpretations and not the exact date, we used all the available satellite data in the year, as we describe next.

D. Sampling Design

We divided the 25 254 reference pixel locations into the reference data set into two subsets for the analysis. We used ∼30% of the pixels (7.577) for calibrating individual random forest classifiers for each year by matching each year's reference class measurement with remote sensing reflectance values from all available Landsat images in the year. We also kept the remaining sampling units for prediction (called "prediction data set" hereon). Therefore, each observation consists of six reflectance spectral bands: blue, green, red, infrared, two near-infrared bands, and a reference class value. Because we used all available images in the period, one or more Landsat multispectral records may be used for each pixel in a single year, all labeled using the year's reference class value of the pixel. The corresponding records were used as "independent" observations for calibration of the classifier in a particular year (see Fig. 2).

We fitted the classifiers in R [34] using the RandomForest [35] package. We fitted the models with the following parameters: ensemble size of 30 trees and node size of five and three variables per split. Tests (not shown here) indicated that the user's, producer's, and total accuracies of the classifications were rather insensitive to the values of these parameters (with a significant decrease in accuracy with node sizes bigger than 5, but no improvement with smaller node sizes). Thus, they were chosen simply to keep computation times manageable.

The year's classifier was then used to obtain predicted class probabilities for all pixels in the prediction sample only. We collapsed all within-year predictions to a single yearly prediction by averaging within-year class membership probabilities (see Fig. 2). The year's hard class predictions for pixels were obtained posteriorly by assigning the class with the highest probability as the pixel label.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASSES MAPPED BY THE MAPBIOMAS PROJECT ACROSS BRAZIL

LEVEL 1	LEVEL 2	LEVEL 3	DESCRIPTION[1]
Forests	Natural Forest	Forest Formation	Vegetation types with a predominance of tree species with high-density continuous canopy, areas that had disturbed by fires and/or logging, and forest resulting from natural
		Savannah	Vegetation types with a tree layer varying in density, distributed over a continuous shrub-herbaceous layer.
		Mangrove	Dense and evergreen forest formation often flooded by the tide and associated with the coastal mangrove ecosystem.
		Secondary Forest	Regrowth after clear-cut deforestation
	Forest Plantation	Forest Plantation	Planted tree species for commercial use.
Non-Forest Natural Formation	Non-Forest Formation in Wetlands		Floodplain with fluvial and lake influence, subject to periodic or permanent flooding, is located along watercourses and in lowlands that accumulate water, with herbaceous shrub vegetation and/or arboreal and pioneer formations, and marshes (marine influence).
	Grasslands		Vegetation type with predominance of herbaceous stratum, including patches with a well-developed shrub-herbaceous stratum.
	Apicum		Hypersaline tidal flats
	Rocky Outcrop		Naturally exposed rocks in the terrestrial surface without soil cover, often with the partial presence of rock vegetation and high slope.
	Other Non-Forest Natural Formation		Natural grasslands, Savanna, Park Savanna, Steppe Savanna, Woody-Grassland Savanna, "Campinarana" in the Amazon biome.
Agropastoralism	Pasture		Pasture areas, natural or planted, related to farming activity. In particular, in the Pampa and Pantanal biomes, part of the area classified as Grassland Formation also includes pasture areas.
	Agriculture	Perennial Crops	Areas predominantly occupied by annual crops and, in some regions (mainly in the Northeast), with perennial crops.
		Semi-Perennial	Areas cultivated with the sugarcane plantation.
		Mixed Crops	A mix of perennial and semi-perennial crops.
	Agriculture & Cattle Ranching		Farming areas where it was not possible to distinguish between pasture and agriculture.
Non-Vegetated Area	Beach or Dune		Sandy areas, with bright white color, where there is no vegetation predominance of vegetation of any kind.
	Urban Infrastructure		Urban areas with a predominance of non-vegetated surfaces, including roads, highways, and constructions.
	Other Non-Vegetated Area		Non-vegetated surface areas (infrastructure, urban areas, or mining) not mapped into their classes, and exposed soil areas (mainly sandy soil) not classified as Grassland
	30. Mining		Areas related to extensive mineral extraction, with clear soil exposure due to heavy machinery. Only areas belonging to DNPM's chart (SIGMINE) are considered.
Water Bodies	River, Lake or Ocean		Rivers, lakes, dams, reservoir, and other water bodies
	Aquaculture		Artificial lakes, where aquaculture and/or salt production activities predominate
Non-Observed			Areas blocked by clouds or atmospheric noise, or with the absence of ground observation masked from the analysis.

[1] From [33]

E. Class Proportion Estimates

After the prediction phase, class proportions in the prediction data set were estimated by: 1) counting the number of pixels classified in each class and dividing by the number of pixels in the prediction data set

$$\widehat{\mathrm{PC}}_c = \frac{\sum_N I_i(\mathrm{class} = c)}{N} \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.

TABLE II
NUMBER OF SAMPLING UNITS (AVERAGES OVER THE YEARS) PER CLASS IN EACH SUBSET

CLASS	TOTAL SAMPLE SIZE	PIXELS USED IN RANDOM FOREST CALIBRATION[1]	NUMBER OF PREDICTED PIXELS[2]	SAMPLE SIZE FOR AREA ADJUSTMENT[3]
Natural Forest	19743	8286	11457	3896
Pasture	2986	1270	1715	609
Other Non-Forest Natural Formation	1593	666	927	311
River, Lake or Ocean	497	203	294	99
Non-Observed	243	92	151	50
Agriculture	134	54	80	25
Other Non-Vegetated Area	18	8	10	3
Forest Plantation	18	7	11	3
Urban Infrastructure	16	6	10	4
Mining	5	2	3	2

[1]Sample size used for calibrating the random forest model. These were not used in the predictions
[2]Average number of pixels for which prediction was performed using the calibrated random forest.
[3]Sample used as extra reference data for computing post-stratification estimates.

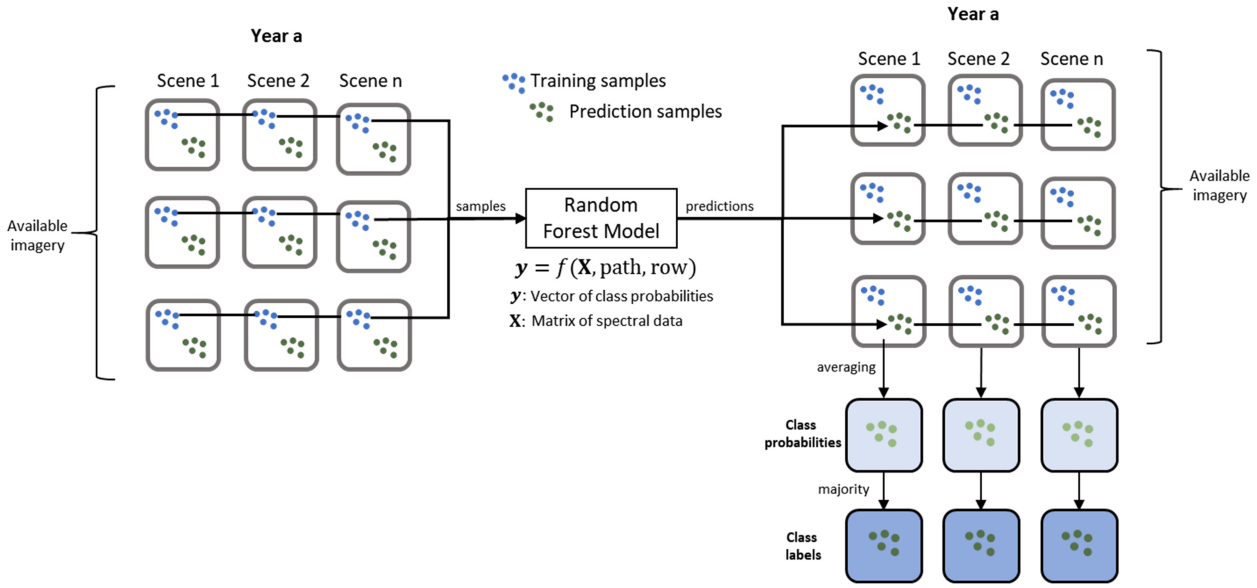


Fig. 2. Sampling frame flowchart. Because multiple imageries are available per scene each year, all of them were used to model the random forests. We have then included variables for the path and row of each scene in the random forests to prevent selection bias. The prediction set was built using the random forest classifier to predict probabilities for each sample, again using all available imagery. This resulted in multiple predictions per year for each prediction sample, which were averaged to arrive at a single set of class probabilities per sample, per year, as the image indicates. Finally, label predictions were obtained by maximum probability assignment (i.e., the majority) using the (averaged) class probabilities.

where N is the number of pixels in the population and 2) by averaging the class membership probabilities estimates of each class over all pixels

$$\widehat{PP_c} = \frac{\sum_N P_i(\text{class} = c)}{N}. \tag{4}$$

The class proportion estimates (computed for the prediction sample) using class pixel counting and the probabilities were compared visually and by their overall relative error, which is the simple average of the yearly class error $(\hat{P}_c - P_c)/P_c$, where $\hat{P}_c$ is the corresponding estimate.

Besides, a comparison with wall-to-wall poststratification area proportion estimates using additional reference data was included, as described in the following.

F. Poststratification Population Class Proportion Estimates Using Extra Reference Data

We also evaluate our predictions by comparing them with the original *Mapbiomas*' *collection 4.0* area estimates for the Amazon, uncorrected and corrected using additional 5000 extra reference pixels, chosen by simple random sampling of the validation data set. If the class proportion estimates using membership probabilities are consistent, they should be comparable to poststratification wall-to-wall estimates of the area using this additional reference data. The poststratification (corrected) population class proportion estimates are given by

$$\widehat{\text{PS}_c} = \sum_k \left\{ \frac{\sum_n (R_{ci} \times M_{ki})}{\sum_n M_{ki}} \times \frac{\sum_N M_{kj}}{N} \right\} \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.

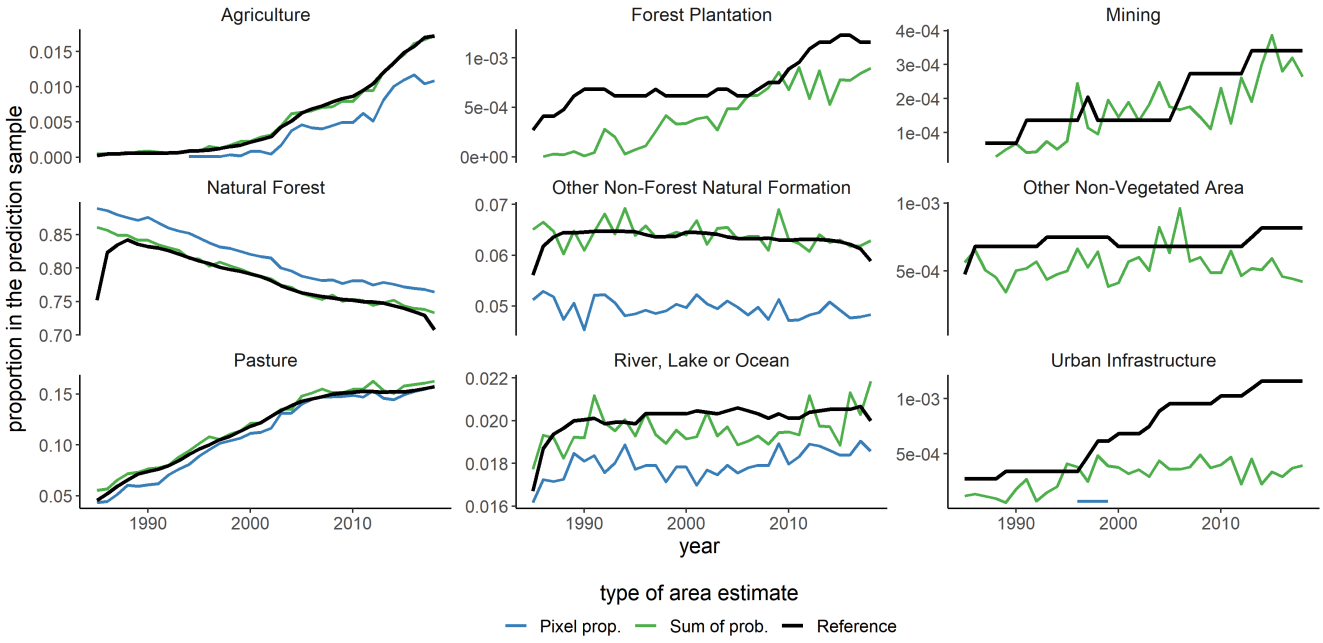


Fig. 3. Comparison of the time series of class proportion estimates based on pixel counting (light-gray lines) based on class membership probabilities (dark-gray lines), and reference class proportions (black lines) in the prediction sample. Each subplot shows the time series of a class in the second taxonomic level of classes mapped by the Mapbiomas project.

where $\widehat{\mathrm{PS}_c}$ is the estimate of P_c, the proportion of the area of reference class c in the population, and R_c and M_k, l, $k = 1, \ldots, K$, are indicator variables representing the reference and predicted (or mapped, or classified, or labeled) classes, respectively. The indices $i = 1, \ldots, n$ identify the pixels in the sample and $j = 1, \ldots, N$ the pixels in the population. These are equivalent to the poststratification estimates currently recommended in the literature [20] or simple random sampling [15]. Their standard errors are also estimated, using

$$\mathrm{Var}\left[\widehat{\mathrm{PS}_c}\right] = \sum_k \left(\frac{n}{N}\right)^2 \left(\frac{\sum_N M_{kj}}{\sum_n M_{ki}}\right)^2 \frac{s_k^2}{n} \tag{6}$$

where $s_k^2 = (\sum_n (R_{li} - (\sum_n R_{ci} / \sum_n M_{ki}) M_{ki})^2)/(n-1)$, k, i, and j as above.

These are unbiased estimates of the class area proportions, but they require additional data. We will show that the random forest's predicted membership probabilities can be used to calculate areas and discuss conditions that circumvent the need for poststratification correction for computing unbiased area estimates.

Equations (4) and (5) can be viewed as design- and model-based estimates of the total, respectively. In this sense, the consistency of the probabilities-based total area estimates of (4) is also conditioned on the sample design. For details on the conditions for valid model-based estimation of finite population parameters, as well as a complete discussion on how to build models for valid model-based inference for different sampling designs, see [36].

G. Consistency of Area Estimates

We performed one last analysis to assess the consistency of the area estimates based on the probabilities. We ran additional random forest models using 1000, 2000, 3000, and 5000 along with the model using all the 7577 calibration sampling units to observe the convergence of the class proportion estimates to the reference proportions in the prediction data set.

III. Results

A. Class Proportion Estimates for the Prediction Sample

Fig. 3 compares the class proportions estimates based on pixel-counting (light-gray lines) and membership probabilities (dark-gray lines) and reference class proportions (black lines) for each year, for the prediction set of 15 000 sampling units. As expected, the proportions based on pixel-counting tend to overestimate common classes and underestimate the rarer ones. In contrast, the estimates based on the probabilities are closer to the true proportions for all classes.

The relative overall error (see Table III) was approximately 40% points lower for the proportion estimates based on probabilities than for the pixel-counting predictions. At the same time, for individual classes, the improvement was at least 3% points. The gain was largest for less abundant classes.

B. Corrected Class Proportion Estimates for the Population

The comparison with class proportion estimates for Mapbiomas' collection 4 land cover mapping is shown in Fig. 4. The envelopes shown in the figure are the population 95% confidence intervals based on standard errors given by (6). The uncorrected population class proportion estimates based on simple pixel-counting of Mapbioma's classifications (dashed lines) and the proportions estimated using the probabilities (solid lines) in the prediction sample subset are also shown.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.

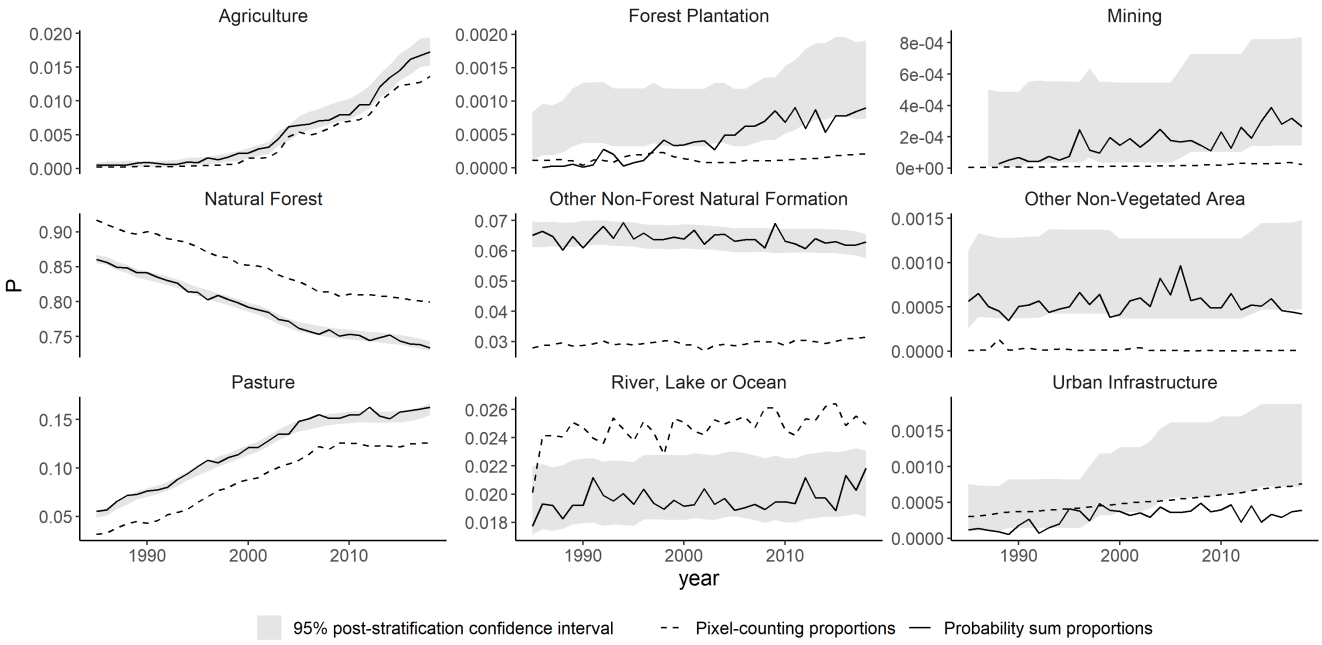


Fig. 4. Population poststratification 95% confidence intervals for population class area proportions (gray envelopes), along with population pixel-counting class proportions (dashed lines) and sample-based probability-based estimates (black lines) using the prediction set of 15k pixels. Class pixel counts were obtained from the wall-to-wall mapping done by Mapbiomas collection 4. Poststratification estimates use extra 5000 reference points to correct pixel counting estimates with (5), while estimates based on the probability are estimated using just the predictions for the 15k prediction set due to computational limitations. Regardless, the probability-based estimates calculated for the sample agree with the population poststratification estimates.

TABLE III

AVERAGE RELATIVE ERROR OF ESTIMATED SAMPLE CLASS PROPORTIONS BASED ON PIXEL COUNTING AND MEMBERSHIP PROBABILITIES, GIVEN BY $(\hat{P}_c - P_c)/P_c$ [SEE (1)–(3)]

CLASS	PREDICTION TYPE	
	Pixel Counting	Membership Probabilities
Agriculture	-65.9%	9.3%
Forest Plantation	-100.0%	-50.2%
Mining	-100.0%	-17.4%
Natural Forest	4.7%	1.0%
Other Non-Forest Natural Formation	-21.7%	1.3%
Other Non-Vegetated Area	-99.8%	-22.7%
Pasture	-6.0%	3.6%
River, Lake or Ocean	-10.5%	-2.3%
Urban Infrastructure	-99.1%	-52.9%
Average over all classes	-55.4%	-14.5%

Note that we have not produced estimates based on the probabilities for every pixel in the biome but only for the prediction sample. Unfortunately, the Google Earth Engine does not provide the class probabilities along with predictions for a random forest classification if there are more than two classes. Due to this constraint, we made predictions to our sample only for computational limitations. Nevertheless, our results support the superiority of class proportion estimates based on membership probabilities in relation to both the uncorrected pixel counting proportions and the corrected poststratification estimates. This is expected since these area estimates use the class membership probabilities conditioned on the covariates (spectral bands), as opposed to the poststratification estimates, which uses the map categories only as strata. The claim is also substantiated by noting that the area estimates based on the probabilities (see Fig. 3) fall within the confidence bounds of the corrected area estimates even for the most frequent classes, for which the debiased estimates have the highest precision. The estimates based on the membership probabilities (solid black lines in Fig. 4 above) are most accurate for the classes with more predicted pixels.

Also, the class areas reported by counting pixels in the maps (dashed lines in Fig. 4) fail to depict some interesting temporal patterns, such as the slightly earlier delay in pasture expansions stabilization and a potential expansion of mining and forest plantation in the Amazon. These patterns are more clearly captured by the estimates corrected by the reference data and by the ones based on the membership probabilities.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.

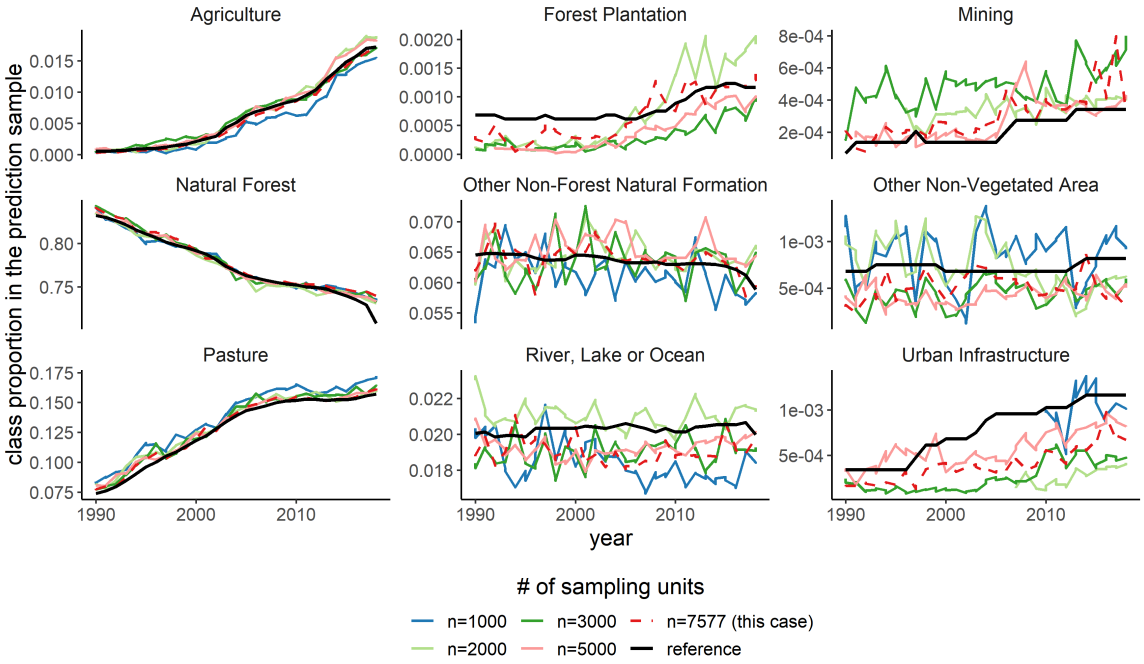


Fig. 5. Class proportion estimates for different sample sizes used to calibrate the Random Forest models.

Though the results above show the superiority of probability-based area predictions, in this particular case, the estimates could still be somewhat biased because, as mentioned in Section II-D, the actual sampling strategy was not simple random sampling. We believe that this bias is small in this case, given the small variability of the sampling weights in the region. Even so, any bias would only impact the wall-to-wall samples because our results solely based on samples are balanced by the design, as we sampled the full sample samples by simple random sampling.

C. Consistency of Area Estimates Based on the Membership Probabilities

Fig. 5 depicts the consistency property of the class proportions estimates based on the probabilities. As the sample size increases, the error reduces around the reference class proportions. Estimates based on 5000 sampling units were similar to the ones obtained with 7500 sampling units for this study. However, while the class proportion estimates of the biggest classes, such as natural forests or pastures, seem to be adequately approximated even using 3000 or 2000 sampling units, estimates for the smaller classes, such as urban infrastructure and other nonvegetation mining, could benefit from more calibration sites. However, estimates could be better if the approach is applied to the entire map because more pixels would be involved in the area estimate.

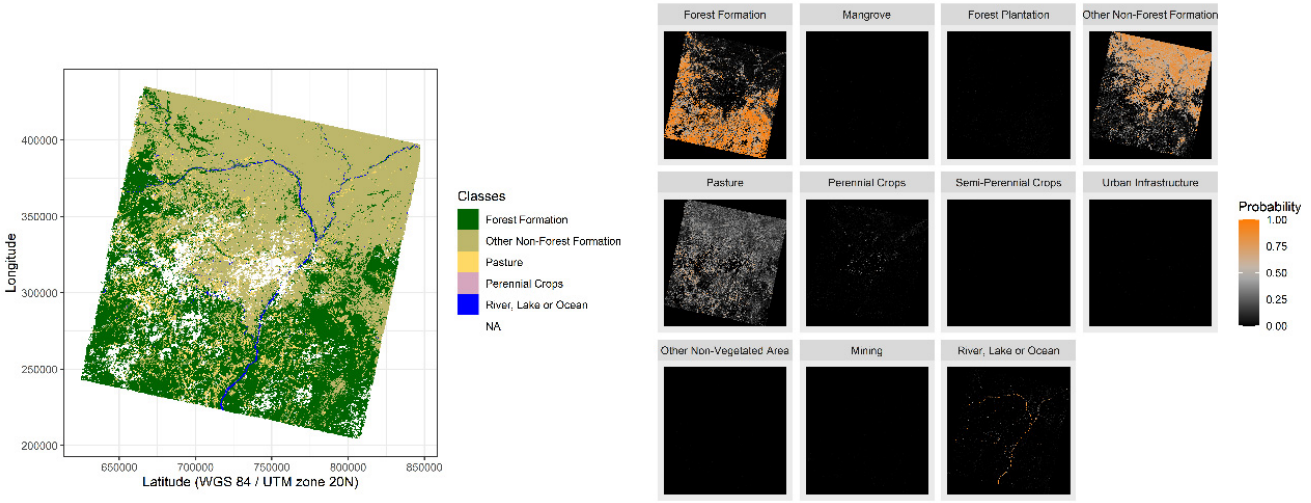
Strangely, the estimates for rivers, lakes, and oceans seem to be converging to a different value, which could indicate a lack of explanatory power of the models for that class, possibly due to the high seasonality of water bodies in the region [37].

An example (full wall-to-wall estimation for the study region is computationally prohibitive due to limitations in this research) comparison example between the classification and probabilities is shown in Fig. 6. We observe that the area of (natural) pastures, perennial crops, and water bodies could be highly underestimated by the hard classification. On the other hand, other nonforest formations could be overestimated in the northern part of the scene due to its confusion with pastures and forests, and the distribution of perennial crops could be wider than what can be observed in the hard classification.

IV. Discussion

The results above reinforce the value of retaining and using the class membership probabilities from a random forest classification (or any weighted nearest neighbor classifier). Besides being useful for displaying spatial uncertainty, both theory and empirical results show that a consistent estimate of the class areas can be obtained using them, and this can be done to assess long-term annual area trends for many land cover classes. Our results showed that more accurate estimates are obtained for the larger classes, but estimates for smaller classes could improve if the approach were applied to the entire map. In that case, many more pixels would be involved in the area proportion estimates, implying that estimates from the sample data will tend to be closer to the population parameter, by the law of large numbers [38]. In simpler words, if the remote sensing data over the domain are free from clouds and other kinds of impurities, averaging probabilities for the entire map would produce more precise class proportion estimates, decreasing the need for additional sampling units to remove bias. This argument is supported by the fact that random forests produce probabilities conditioned on all spectral variables, while the poststratification estimates are just ratio estimates that use the map classes as strata. Therefore, in the latter case, a much larger sample size

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.



(a) **(b)**

Fig. 6. (a) "Hard" classification and (b) class membership probabilities, constructed from all available Landsat 8 imagery for path/row 232/58 over the year of 2018.

than that used to calibrate the classification calibration would be necessary to improve the estimates based on the class membership probabilities.

We found that using the predicted probabilities from a random forest model may dispense the use of poststratification, but, sometimes, correction is still essential. One example is the apparent case when not the probabilities but only the LULC map is available; then, poststratification is the only way to obtain unbiased estimates. Poststratification may also be used to estimate LULC class areas for classes not included in the original classification. In this case, the original classification map can potentially be used as strata to increase the precision of the estimates that would otherwise use only sample data [39].

Another reason to store the probabilities is that the class membership probabilities from the classifications are valuable for visualization, as the hard class predictions may hide the true spatial distribution of land cover classes, especially for map users interested in the spatial patterns of the most infrequent classes, as a pixel is only labeled a class when its probability is the highest. For the same reason, the probabilities could be used to obtain better estimates for subregions of the map, for which estimates based on the probabilities could wrongly point total absence of the class.

The wall-to-wall LULC class proportion estimates shown in Fig. 4 suggest that the class proportions reported using pixel counting can be considered biased, even when postclassification tuning to fix imperfections of the classifications is used, as is the case of the Mapbiomas project. This reinforces the importance of keeping and using the probabilities for estimating LULC class areas by any land cover mapping initiatives. Mapping membership probabilities can also be useful for directly estimating LULC class area proportions over subregions of the map, such as geographical regions or pixel blocks. It should then be assumed that the random forest model is appropriate for the targeted region, and the subregion should be large so that many pixels are included for the area estimation of each class. This would, perhaps, be the greatest advantage of using the probabilities over poststratification estimates (in the absence of extra abundant high-quality data) because this can be done without the need for additional sampling units or intricate sampling designs. Nevertheless, future research should provide a more detailed characterization of convergent rates and test or derive ways for efficient estimation of the prediction error for global and local estimation of LULC class areas.

Our case study assumes simple random sampling. However, stratified random sampling is a design often used in land cover classification applications. If the sample is stratified proportionally to class size, all weights should be the same, and no adaptations are needed. In fact, our reference data was collected by stratified random sampling proportional to size. If the sampling is not proportional to class size, then the bootstrap selection probabilities must be corrected to ensure that the class probabilities are still consistent [40].

SU is another method that could be used to estimate class areas with reduced bias by the sum of estimated subpixel class areas. In terms of classification accuracy, SU has been outperformed by random forest. An interesting additional question is whether area estimation based on random forest class probabilities would also outperform SU in terms of convergence rates, which should be explored in future work.

We have not discussed how well the probability estimates the area change of LCLU over consecutive years because testing this empirically would require many more sampling units. In theory, transitions could be estimated by modeling transition probabilities directly, which could require a much larger calibration sample or approximated using the yearly estimated class probabilities, assuming independence. The adequacy of this approximation could be the subject of future work.

Another possible topic for future work is whether additional reference data could still be used to improve area estimates

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.

based on random forest class probabilities, as well as what would be an optimal sampling strategy in such a case.

V. Conclusion

We have empirically demonstrated that the class membership probabilities from a random forest model, calibrated using a simple random sampling, can be used to provide consistent estimates of LULC class areas. The method can now be used as a valid alternative to poststratification estimates to obtain estimates of each LCLU class area. The estimate's convergence rates will depend on the discriminatory power of the spectral information used in the model and the size of each class. The area estimates obtained in this way may dispense the need for extra samples to adjust the class areas, which can benefit large-scale land cover mapping projects. For example, the probabilities can also be used to estimate areas in subregions of the map without correction, assuming that the model is appropriate for the subregion, and the subregion includes many pixels. Therefore, land cover and land use mapping projects should retain and, if possible, publish the mapped class membership probabilities along with the mapped LULC class labels. Future work should focus on a more detailed investigation of convergence rates, the relation to the discriminatory power of the models, and on efficient ways of estimating the error variance of the estimates, especially for subregions of the map, which may need to take into account the spatial (cross-)autocorrelation structure of the pixel probability estimates.

References

[1] P. Verburg and T. Veldkamp. (2004). *Land Use Change Modelling: Current Practice and Research Priorities*. Accessed: Mar. 11, 2014. [Online]. Available:http://ftp.wsl.ch/pub/kienast/mast_old/class_no1/Course_material_clue.pdf

[2] J. A. Foley, "Global consequences of land use," *Science*, vol. 309, no. 5734, pp. 570–574, Jul. 2005, doi: 10.1126/science.1111772.

[3] M. Cerreta and P. De Toro, "Urbanization suitability maps: A dynamic spatial decision support system for sustainable land use," *Earth Syst. Dyn.*, vol. 3, no. 2, pp. 157–171, Nov. 2012, doi: 10.5194/esd-3-157-2012.

[4] E. Y. Arima, P. Barreto, E. Araújo, and B. Soares-Filho, "Public policies can reduce tropical deforestation: Lessons and challenges from Brazil," *Land Use Policy*, vol. 41, pp. 465–473, Nov. 2014, doi: 10.1016/j.landusepol.2014.06.026.

[5] J. B. D. S. F. Filho and M. Horridge, "Ethanol expansion and indirect land use change in Brazil," *Land Use Policy*, vol. 36, pp. 595–604, Jan. 2014, doi: 10.1016/j.landusepol.2013.10.015.

[6] A. K. Skidmore, W. Bijker, K. Schmidt, and L. Kumar, "Use of remote sensing and GIS for sustainable land management," *ITC J.*, vol. 3, no. 4, pp. 302–315, 1997.

[7] J. T. Kerr and M. Ostrovsky, "From space to species: Ecological applications for remote sensing," *Trends Ecol. Evol.*, vol. 18, no. 6, pp. 299–305, Jun. 2003, doi: 10.1016/S0169-5347(03)00071-5.

[8] M. Herold, P. Mayaux, C. E. Woodcock, A. Baccini, and C. Schmullius, "Some challenges in global land cover mapping: An assessment of agreement and accuracy in existing 1 km datasets," *Remote Sens. Environ.*, vol. 112, no. 5, pp. 2538–2556, May 2008, doi: 10.1016/j.rse.2007.11.013.

[9] F. Waldner and P. Defourny, "Where can pixel counting area estimates meet user-defined accuracy requirements?" *Int. J. Appl. Earth Observ. Geoinf.*, vol. 60, pp. 1–10, Aug. 2017, doi: 10.1016/j.jag.2017.03.014.

[10] J. D. Malley, J. Kruppa, A. Dasgupta, K. G. Malley, and A. Ziegler, "Probability machines," *Methods Inf. Med.*, vol. 51, no. 1, pp. 74–81, Jan. 2012, doi: 10.3414/ME00-01-0052.

[11] C. Quintano, A. Fernández-Manso, Y. E. Shimabukuro, and G. Pereira, "Spectral unmixing," *Int. J. Remote Sens.*, vol. 33, no. 17, pp. 5307–5340, Sep. 2012, doi: 10.1080/01431161.2012.661095.

[12] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[13] P. O. Gislason, J. A. Benediktsson, and J. R. Sveinsson, "Random forests for land cover classification," *Pattern Recognit. Lett.*, vol. 27, no. 4, pp. 294–300, Mar. 2006, doi: 10.1016/j.patrec.2005.08.011.

[14] V. F. Rodriguez-Galiano, B. Ghimire, J. Rogan, M. Chica-Olmo, and J. P. Rigol-Sanchez, "An assessment of the effectiveness of a random forest classifier for land-cover classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 67, pp. 93–104, Jan. 2012, doi: 10.1016/j.isprsjprs.2011.11.002.

[15] F. J. Gallego, "Remote sensing and land cover area estimation," *Int. J. Remote Sens.*, vol. 25, no. 15, pp. 3019–3047, Aug. 2004, doi: 10.1080/01431160310001619607.

[16] P. Olofsson, G. M. Foody, S. V. Stehman, and C. E. Woodcock, "Making better use of accuracy data in land change studies: Estimating accuracy and area and quantifying uncertainty using stratified estimation," *Remote Sens. Environ.*, vol. 129, pp. 122–131, Feb. 2013, doi: 10.1016/j.rse.2012.10.031.

[17] F. Canters, "Evaluating the uncertainty of area estimates derived from fuuy land-cover classification," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 4, pp. 403–414, 1997.

[18] S. de Bruin, "Predicting the areal extent of land-cover types using classified imagery and geostatistics," *Remote Sens. Environ.*, vol. 74, no. 3, pp. 387–396, Dec. 2000, doi: 10.1016/S0034-4257(00)00132-2.

[19] S. V. Stehman, "Impact of sample size allocation when using stratified random sampling to estimate accuracy and area of land-cover change," *Remote Sens. Lett.*, vol. 3, no. 2, pp. 111–120, Mar. 2012, doi: 10.1080/01431161.2010.541950.

[20] P. Olofsson, G. M. Foody, M. Herold, S. V. Stehman, C. E. Woodcock, and M. A. Wulder, "Good practices for estimating area and assessing accuracy of land change," *Remote Sens. Environ.*, vol. 148, pp. 42–57, May 2014, doi: 10.1016/j.rse.2014.02.015.

[21] W. G. Cochran, *Sampling Techniques*, 3rd ed. Hoboken, NJ, USA: Wiley, 1977.

[22] C. Yang, J. H. Everitt, R. S. Fletcher, and D. Murden, "Using high resolution QuickBird imagery for crop identification and area estimation," *Geocarto Int.*, vol. 22, no. 3, pp. 219–233, Sep. 2007, doi: 10.1080/10106040701204412.

[23] B. D. Wardlow and S. L. Egbert, "A comparison of MODIS 250-m EVI and NDVI data for crop mapping: A case study for southwest kansas," *Int. J. Remote Sens.*, vol. 31, no. 3, pp. 805–830, Feb. 2010, doi: 10.1080/01431160902897858.

[24] H. Müller, P. Rufin, P. Griffiths, A. J. B. Siqueira, and P. Hostert, "Mining dense landsat time series for separating cropland and pasture in a heterogeneous Brazilian savanna landscape," *Remote Sens. Environ.*, vol. 156, pp. 490–499, Jan. 2015, doi: 10.1016/j.rse.2014.10.014.

[25] A. Alam, M. S. Bhat, and M. Maheen, "Using landsat satellite data for assessing the land use and land cover change in kashmir valley," *GeoJournal*, vol. 85, no. 6, pp. 1529–1543, Dec. 2020, doi: 10.1007/s10708-019-10037-x.

[26] S. Baeza and J. M. Paruelo, "Land use/land cover change (2000–2014) in the Rio de la Plata grasslands: An analysis based on MODIS NDVI time series," *Remote Sens.*, vol. 12, no. 3, p. 381, Jan. 2020, doi: 10.3390/rs12030381.

[27] G. Biau, "Analysis of a random forests model," *J. Mach. Learn. Res.*, vol. 13, pp. 1063–1095, Apr. 2012.

[28] C. Zhang and Y. Ma, Eds., *Ensemble Machine Learning*. New York, NY, USA: Springer, 2012.

[29] G. Biau and L. Devroye, "On the layered nearest neighbour estimate, the bagged nearest neighbour estimate and the random forest method in regression and classification," *J. Multivariate Anal.*, vol. 101, no. 10, pp. 2499–2518, Nov. 2010.

[30] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, and R. Moore, "Google Earth engine: Planetary-scale geospatial analysis for everyone," *Remote Sens. Environ.*, vol. 202, pp. 18–27, Dec. 2017, doi: 10.1016/j.rse.2017.06.031.

[31] T. Azevedo, C. M. Souza, J. Shimbo, and A. Alencar, "MapBiomas initiative: Mapping annual land cover and land use changes in Brazil from 1985 to 2017," *AGU Fall Meet. Abstr.*, vol. 2018, p. B22A-04, Dec. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.

[32] C. Souza and T. Azevedo. (Aug. 2018). *MapBiomas General 'Handbook'. Algorithm Theoretical Basis Document (ATBD). Collection 4*. Mapbiomas, 1.0. Accessed: Sep. 23, 2019. [Online]. Available: https://storage.googleapis.com/mapbiomas/mapbiomas-br/v4/ATBDs/ATBD%20Collection%204%20v1.pdf

[33] C. M. Souza *et al.*, "Reconstructing three decades of land use and land cover changes in Brazilian biomes with Landsat archive and Earth engine," *Remote Sens.*, vol. 12, no. 17, p. 2735, Aug. 2020.

[34] R Core Team, *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing, 2018.

[35] A. Liaw and M. Wiener, "Classification and regression by randomforest," *R News*, vol. 2, no. 3, pp. 18–22, 2002.

[36] S. K. Sterba, "Alternative model-based and design-based frameworks for inference from samples to populations: From polarization to integration," *Multivariate Behav. Res.*, vol. 44, no. 6, pp. 711–740, Nov. 2009, doi: 10.1080/00273170903333574.

[37] C. Souza, F. Kirchhoff, B. Oliveira, J. Ribeiro, and M. Sales, "Long-term annual surface water change in the Brazilian Amazon biome: Potential links with deforestation, infrastructure development and climate change," *Water*, vol. 11, no. 3, p. 566, Mar. 2019, doi: 10.3390/w11030566.

[38] J. R. Blum and M. Brennan, "On the strong law of large numbers for dependent random variables," *Isr. J. Math.*, vol. 37, no. 3, pp. 241–245, Sep. 1980, doi: 10.1007/BF02760965.

[39] S. V. Stehman, "Estimating area and map accuracy for stratified random sampling when the strata are different from the map classes," *Int. J. Remote Sens.*, vol. 35, no. 13, pp. 4923–4939, Jul. 2014, doi: 10.1080/01431161.2014.930207.

[40] M. Zhu *et al.*, "Class weights random forest algorithm for processing class imbalanced medical data," *IEEE Access*, vol. 6, pp. 4641–4652, 2018, doi: 10.1109/ACCESS.2018.2789428.



Marcio H. R. Sales was born in Belém, Brazil, in 1979. He received the B.S.S. and Specialization degrees in statistics from the Universidade Federal do Pará, Belém, in 2002 and 2003, respectively, and the M.A.A. degree in geography from the University of California at Santa Barbara, Santa Barbara, CA, USA, in 2007. He is pursuing the Ph.D. degree with the Production Ecology and Resource Conservation Graduate School, Wageningen University & Research, Wageningen, The Netherlands.

From 2002 to 2012, he was an Assistant Researcher with the Instituto do Homem e do Meio Ambiente da Amazonia (IMAZON), Belém. His research interests are spatial statistics and spatiotemporal statistical modeling of the landscape from remote sensing data.



Sytze de Bruin received the M.Sc. degree in soil science and the Ph.D. degree in geoinformation science from the Wageningen University & Research, Wageningen, The Netherlands, in 1987 and 2000, respectively.

He has (co)authored over 70 journal articles, conference papers, and book chapters. His research and education focus on developing and using methods for transforming spatial data into useful geoinformation. He is particularly interested in the assessment of fitness-for-purpose, data acquisition, including spatial and temporal sampling and sensing, and other quantitative methods used for spatial and temporal analyses. He has been involved in several research projects with applications ranging from global land degradation assessment to precision agriculture.

Dr. de Bruin has served as an Associate Editor for the *International Journal of Geographical Information Science* for over a decade.



Carlos Souza, Jr. received the bachelor's degree in geology from the Federal University of Pará, Belém, Brazil, in 1991, the M.Sc. degree in soil science from Penn State University, State College, PA, USA, in 1997, and the Ph.D. degree in geography from the University of California at Santa Barbara, Santa Barbara, CA, USA, in 2005.

He is an Associate Researcher with the Instituto do Homem e do Meio Ambiente da Amazonia (IMAZON), Belém, focusing on the field of remote sensing for mapping and monitoring forests, forest degradation, land use and land cover change, surface water monitoring, and spatial modeling. He has (co)authored more than 100 journal articles, conference papers, and book chapters.

Dr. Souza, Jr. received the Skoll Award on Social Entrepreneurship in 2010 and the Conservation Fellowship from the Mulago Foundation in 2017.



Martin Herold was born in Leipzig, Germany, in 1975. He received the Ph.D. degree from the University of California at Santa Barbara, Santa Barbara, CA, USA, in 2004, and the Habilitation (second German Ph.D.) degree in operational global land cover observation and assessments in 2009.

He is a Professor of geoinformation science and remote sensing with the Wageningen University & Research, Wageningen, The Netherlands. He coleads the land cover team Global Observations of Forest Cover and Land Dynamics (GOFC-GOLD) and coordinates the research and development component of the Global Forest Observations Initiative (GFOI). He is an expert in monitoring forest and land use change and related biomass and carbon emissions, global land cover mapping and assessments, integration of ground and remote sensing data, international coordination and standardization, and accuracy assessment procedures and uncertainty management for remote sensing data products and uptake in policy and applications. He is active in the development of monitoring systems in the context of United Nations Framework on Climate Change (UNFCCC) and the Sustainable Development Goals. He has published more than 150 scientific articles. He has been supervising Ph.D. students. He supports capacity development initiatives for moving innovative satellite and ground-based approaches into practice.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:03:07 UTC from IEEE Xplore. Restrictions apply.